

Effective Learning Behavior of Students' Internet Based on Data Mining

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Abstract: With the rapid development of information technology and the widespread application of network technology, it has had a great impact on traditional learning methods. Teaching and learning based on the network environment has become an important form and part of school teaching. As a new type of learning method, online learning has its unique advantages and potential for development. It breaks through the limitations of time and space in traditional teaching methods, and brings great convenience and freedom to learners, but there are also many problem. The purpose of this article is to analyze the effective learning behavior of students on the Internet based on data mining. This article analyzes learning behavior and builds a learning behavior mining model on this basis. Use the improved algorithm to mine and analyze the learning behavior data, including preprocessing and conversion of the data in the database, respectively mining the relationship between the learning sequence pattern and the learning behavior and effect, and finally the mining results explanation and evaluation. This paper does certain research and improvement on association rules and sequential pattern mining algorithms, and on this basis, uses the improved algorithm to collect, mine and analyze learning behavior data, and finally apply data mining technology to personalized the learning system optimizes the mechanism of online learning and the learning experience of users. Research on related methods of learning behavior mining, detailed analysis of association rules and sequence pattern mining, and optimization of Apriori algorithm and AprioriAll algorithm. The performance of the two algorithms and their optimized algorithms are compared respectively. The experimental results show that the improved algorithm has better performance improvement.

Key words: Data Mining, Online Learning Behavior, Association Rules, Learning System

I. INTRODUCTION

Learning in the network environment has great advantages [1-2], which is mainly reflected in openness, flexibility, multimedia, autonomy, interactivity and sharing, etc. However, there are common learning models in current online learning single [3-4], simple accumulation of learning resources, imperfect management and monitoring functions, inflexible interaction methods, low intelligence, etc. [5-6], and the learning ability, personal interests and personal interests of learners participating in online learning habits, personal knowledge base, cognitive level, and effort level all have great differences. How to ensure the quality of online learning and make online learning more effective and efficient has become a key issue in online education research [7-8].

In the research of analyzing the effective learning behavior of students on the Internet based on data mining, many scholars have conducted research on it and achieved good results. For

example, Sarwar S conducted an empirical study on the impact of video production decisions on student participation in online education videos to measure student viewing the length of each video, and whether they try to answer the evaluation questions after the video, from a quantitative and qualitative perspective, put forward a series of suggestions to help teachers and video producers make better use of online video formats [9]. Ma Z uses a two-layer hidden Markov chain to build a MOC learner's learning behavior model, so as to extract the learning behavior characteristics of MOOC learners related to the learning effect [10].

This paper has done research on related methods of learning behavior mining, detailed analysis of association rules and sequence pattern mining, and optimized the Apriori algorithm and AprioriAll algorithm. The performance of the two algorithms and their optimized algorithms were compared and tested, and the results showed that the improved algorithm has a better performance improvement.

II. EFFECTIVE LEARNING BEHAVIOR OF STUDENTS' INTERNET BASED ON DATA MINING

A. Effectiveness of Online Learning

At present, most of the analysis of effectiveness in online learning is from a theoretical perspective, and it is difficult to conduct a comprehensive quantitative sentence research. The connotation of the effectiveness of online courses includes effective environment, effective resources, effective design, effective teaching, effective learning, etc. This research mainly discusses the effectiveness from the evaluation of learning, and the evaluation of student learning, that is, the evaluation of student online learning behavior and learning results. Grasp, judgment and evaluation may include specific content such as grades, progress, ability growth, and other achievements and performance.

1) Informatization teaching design

Informatization instructional design examines learners' ability to practice and apply basic knowledge of modern educational technology. Informatization instructional design is also a necessary ability for future teachers. Learners are required to design for the subject they are learning to enhance the support of educational technology to other subjects.

2) Online exam

The online exam examines the learner's mastery of the course content. All questions come from the corresponding knowledge points. The online exam can be conducted anywhere. Students will not have too much psychological burden. They can answer in a relaxed and pleasant way, so online the test

results can best reflect the learners' mastery of knowledge points and the true learning level.

3) Intel Future Practice Collection

The purpose is to allow learners to experience task-driven methods, learner-centered autonomy, cooperation, and exploration, and other advanced learning concepts through the completion of Word documents, Excel spreadsheets, and website production. And to issue certificates to outstanding students, to encourage learners to explore the spirit.

4) Final closed book exam

The closed-book test includes noun explanations, fill-in-the-blank questions, short answer questions, and essay questions. Most of them are subjective and open questions. The questions are flexible and diverse and examine students' comprehensive abilities.

Adopting diversified and prominent process evaluation methods is conducive to improving learners' interest and continuity in learning. The evaluation methods of Guoluo courses are effective, and effective evaluation methods are used to evaluate learners. The evaluation results are convincing. Forceful.

B. Construction of Learning Behavior Analysis Model

1) Learning behavior analysis task

There are generally two main purposes for learning behavior mining and analysis:

a) Through mining and analysis of the learning sequence when students visit the corresponding learning platform, valuable association rules and group feature sequences are obtained, which are used to make adaptive recommendations for learning content, realize personalized teaching, and enable online learners Have a better learning experience and improve learning efficiency.

b) Analyze students' online learning behavior and provide teaching guidance. Through the correlation analysis of online learning-related attributes (including courseware on-demand, online Q&A, logging on to learning platforms, access to learning forums, etc.) and the effect of online learning, the impact of these behavioral factors on learning effects is discovered, so as to facilitate the students Reasonable and efficient teaching guidance, in which the learning effect is generally measured by the results of the course.

2) Analysis model of learning behavior

a) Data storage layer: The data set provides the underlying support and data support services for the learning behavior analysis model. The data in the data set has multiple sources, generally web log data generated by the learner's learning behavior and dynamically generated or released during the learning process behavioral data. Most of these data are in a dispersed state and need to be collected and preprocessed. In order to facilitate the mining and analysis of learning behavior data, a database of learning behavior characteristics is constructed and the preprocessed data is stored in the database.

b) Learning behavior mining analysis layer: It is mainly used to mine and analyze the data from the data set after

preprocessing, and to stubbornly analyze the characteristics of learners and learning patterns.

c) Personalized learning layer: The results of learning behavior mining are finally presented to learners and teachers in a visual manner. Learners can obtain learning resources that meet their needs, and teachers can learn online based on learners' learning characteristics conduct certain guidance on behavior, or make appropriate adjustments to teaching content based on this.

C. Learning Behavior Mining Method

1) Apriori algorithm

a) Search for transactions in the database to generate candidate k-itemsets.

b) According to whether the items in the item set meet the minimum support $\min_sup$, generate frequent k-item sets.

c) Perform connection and pruning operations, generate candidate k+1-itemsets from frequent k-itemsets, and repeat the above steps until the generated candidate k+1-itemsets are empty.

d) In the obtained frequent item sets, use the minimum confidence $\min_conf$ to generate strong association rules.

2) Apriori algorithm improvement

The Apriori_M algorithm is described as follows:

$$k1; M_{c1} = \text{ScanningAndJudge}(D) \quad (1)$$

Scan the database to generate a candidate 1-item matrix M_{c1} , and then according to the formula:

$$L_k = \text{Trim_matrix}(M_{ck}) \quad (2)$$

Process the candidate k-item matrix to obtain frequent k-item sets, and then:

$$M_{ck+1} = M_Gen(M_{fk}) \quad (3)$$

The first step is to scan the database and initialize the candidate 1-item matrix M_{c1} according to the results of the scan. The behavior data items of the matrix and the columns indicate transactions, and the specific Boolean value of the matrix indicates whether the data item is in the transaction of the current column. If it exists, it is 1, otherwise it is recorded as 0,

III. EFFECTIVE LEARNING BEHAVIOR OF STUDENTS' INTERNET BASED ON DATA MINING

A. Algorithm Experiment Comparison

In the process of generating frequent itemsets in experiments, Apriori algorithm needs to continuously scan the database to count the support of itemsets. In the algorithm proposed in this article, only the candidate 1-item matrix needs to be obtained during the first database scan, and then the generation of frequent itemsets only needs to surround this matrix, and there is no need to scan the database, which reduces I/O. The overhead, but need to consume additional storage space. When generating frequent itemsets, both the minimum support $\min_sup$ and the size of the data set will affect the efficiency of the algorithm.

B. Apriori Algorithm Experimental Improvement

The main idea of the algorithm is to judge the occurrence of all candidate 1-itemsets in the transaction and obtain their

Boolean values. If they exist, they are recorded as "1", otherwise, they are recorded as "0", and then stored in the matrix. The corresponding position of to get the candidate 1-item matrix. Then the candidate 1-item matrix is calculated to generate a frequent 1-item matrix, where the first column of the frequent 1-item matrix is the frequent 1-item set. Then the candidate 2-item matrix is generated from the frequent 1-item matrix, and so on, until the frequent k-item set is finally obtained.

IV. EXPERIMENTAL RESEARCH AND ANALYSIS OF THE EFFECTIVE LEARNING BEHAVIOR OF STUDENTS' INTERNET BASED ON DATA MINING

A. Cluster Analysis of Learners' Online Learning Behavior

In order to better compare the algorithm proposed in this article with the original Apriori algorithm, the two algorithms were coded in the Java language on MyEclipse and Mushroom from the UCI machine learning data source was selected as the data set for association rule mining. Let the two algorithms run under different support conditions, and the running time results are shown in Table 1.

Table 1. Comparison of the time efficiency of the two algorithms when the minimum support is different

Minimum support	Apriori algorithm	Apriori_M algorithm
10%	16	8
20%	27	13
30%	38	19
40%	58	27
50%	66	31

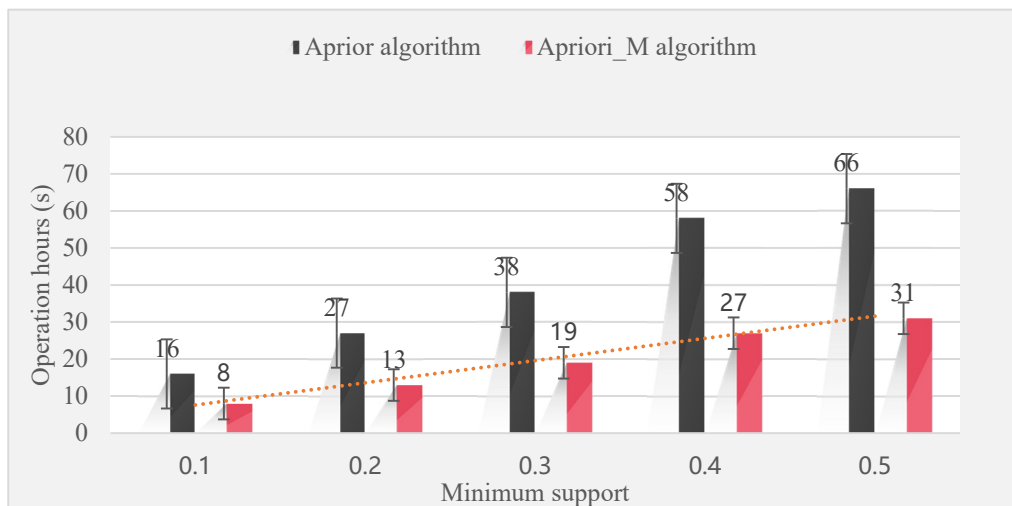


Figure 1. Comparison of the time efficiency of the two algorithms when the minimum support is different

It can be seen from Figure 1 that when the support degree is small, the Apriori_M algorithm improves the efficiency of the Apriori algorithm more significantly. Under different support degrees, the Apriori_M algorithm does not require frequent database scans, so it takes less time than the Apriori algorithm.

B. Learning Times and Learning Frequency

In order to obtain the operating conditions of the algorithm under different transaction data volumes, the minimum support degree is 30%, and 3000, 5000, 7000, and 9000 data records are respectively taken for calculation. Table 2 shows the execution time results of the two algorithms.

Table 2. Comparative analysis of the time efficiency of the two algorithms when the number of transactions is different

Number of transactions (a)	ApriorAll algorithm	Optimized algorithm
3000	10.36	9.21
5000	19.63	17.15
7000	21.41	15.26
9000	28.64	19.29

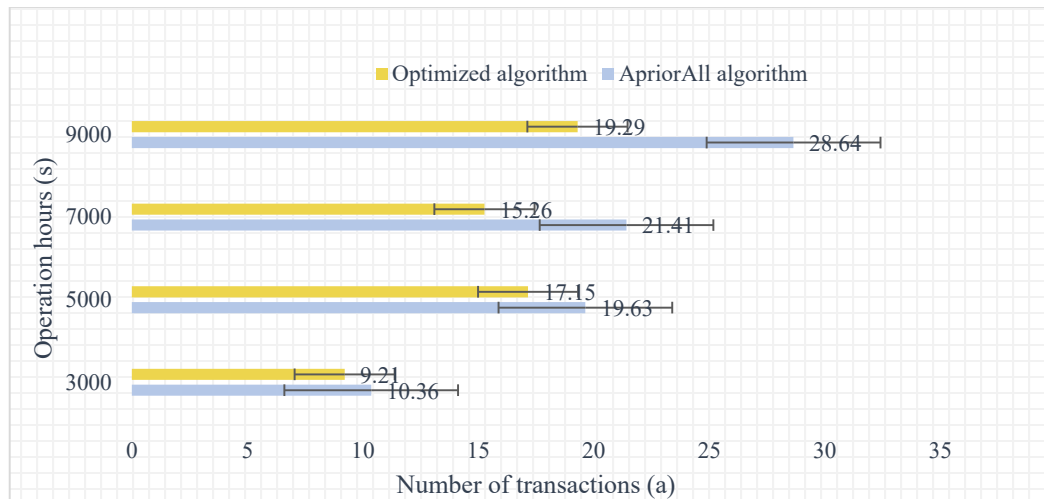


Figure 2. Comparative analysis of the time efficiency of the two algorithms when the number of transactions is different

As shown in Figure 2, when the minimum support $\min_sup$ is the same, as the amount of transaction data increases, the time required by the two algorithms will continue to increase, and there is less data in the initial database. The running time of the Apriori algorithm and the Apriori_M algorithm is similar. It's almost the same, but as the amount of data continues to increase, because the Apriori algorithm needs to continuously scan the database, it takes a lot of time, while Apriori_M only needs to scan the database once, and the efficiency is significantly higher than the Apriori algorithm.

V. CONCLUSIONS

This article first introduces the meaning and background of the topic, and elaborates on the current situation of online learning behavior analysis at home and abroad and the development of data mining technology. This article describes the current development of online learning behavior and data mining technology at home and abroad. After the analysis, the research purpose of this article is put forward. This article introduces and analyzes the association rule algorithm and sequence pattern algorithm in data mining in detail, including related concepts and the process of classic algorithms. And based on the analysis of the original algorithm, a certain optimization is carried out. Finally, the performance of the two algorithms is compared and tested. This paper constructs a network learning behavior mining model and uses the improved association rule mining method to analyze the learning behavior, which mainly includes the preprocessing of the data in the data, the mining of the learning sequence pattern and the association between the learning behavior and the effect Mining relationships, and interpreting and evaluating the mining results.

References

- [1] Gksu I , Bünyamin ATICI. Learners' Evaluation Based on Data Mining in a Web Based Learning Environment[J]. Journal of Computer and Education Research, 2015, 3(5):78-95.
- [2] Venkatraman S , Alazab M , Vinayakumar R . A hybrid deep learning image-based analysis for effective malware detection[J]. Information Security Technical Report, 2019, 47(Aug.):377-389.
- [3] Xiao J , Wang L , Zhao J , et al. Research on Adaptive Learning Prediction Based on XAPI[J]. International Journal of Information and Education Technology, 2020, 10(9):679-684.
- [4] Hou R , Kong Y Q , Cai B , et al. Unstructured big data analysis algorithm and simulation of Internet of Things based on machine learning[J]. Neural Computing and Applications, 2020, 32(10):5399-5407.
- [5] Kamalakkannan S , Prasanna S . Discovering the behavior of the students using data mining techniques[J]. International Journal of Engineering & Technology, 2018, 7(2):518-521.
- [6] Cao K , Wang G , Han D , et al. Classification of Uncertain Data Streams Based on Extreme Learning Machine[J]. Cognitive Computation, 2015, 7(1):150-160.
- [7] Zhao Y . Research on wireless distributed financial risk data stream mining based on dual privacy protection[J]. EURASIP Journal on Wireless Communications and Networking, 2020, 2020(1):1-12.
- [8] Lu H , Setiono R , Liu H . Effective data mining using neural networks[J]. Knowledge & Data Engineering IEEE Transactions on, 2016, 8(6):957-961.
- [9] Sarwar S , Qayyum Z U , Safyan M , et al. Graphs Resemblance based Software Birthmarks through Data Mining for Piracy Control[J]. Proceedings of the Institute for System Programming of RAS, 2019, 31(2):171-186.
- [10] Ma Z , Huang L , Jin Z . Ecological environment monitoring of mining area based on internet of things[J]. IPPTA: Quarterly Journal of Indian Pulp and Paper Technical Association, 2018, 30(5):423-428.